

Nuclear gluonometry with a tensor-polarized ${}^6\text{Li}$ beam at the Electron-Ion Collider: projected inclusive and coherent $\cos 2\phi$ measurements

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Abstract. The double-helicity-flip structure function $\Delta(x, Q^2)$ of a spin-1 target is generated at leading twist by gluons alone and has never been measured. We project its measurement at the Electron-Ion Collider with a transversely tensor-polarized ${}^6\text{Li}$ beam, in inclusive deep inelastic scattering, where only the scattered electron is detected, and in coherent diffraction with the ${}^6\text{Li}$ recoiling intact. With the leading-order bag-model moment of Sather and Schmidt ported to ${}^6\text{Li}$ as a labelled scenario, the inclusive $\cos 2\phi$ amplitudes at the four most sensitive (x, Q^2) bins, $x = 0.011\text{--}0.14$ and $Q^2 = 1.1\text{--}14 \text{ GeV}^2$, are $(0.44\text{--}0.95)\times 10^{-2}$ at $P_{zz} = 0.6$, against statistical errors of $(1.4\text{--}4.5)\times 10^{-4}$ in one EIC year (10 fb^{-1} per nucleon): a $21\text{--}44\sigma$ measurement per bin whose x dependence discriminates between moment-constrained interpretations, and an extraction of $x\Delta(x, Q^2)$ itself to $2\text{--}9\%$ in the best bins. At the reconstructed level, with the spin-state-sorted ratio of $m = \pm 1$ -rich and $m = 0$ -rich bunches and a hadronic-method y from a PYTHIA 8 final state, the errors are $(0.9\text{--}3.0)\times 10^{-4}$ and the amplitude is unbiased. The coherent channel carries the same modulation with two separable mechanisms — a nuclear-shape term anchored on the polarized-deuteron colour-glass calculation, predicted to flip sign for ${}^6\text{Li}$, and a flat exotic-gluon term — and ${}^6\text{Li}$'s anomalously small quadrupole moment makes it a null test. Its intact recoil keeps the beam rigidity and, with the Yellow Report's beam divergences, cannot be tagged at IP6 with any published optics; the channel defines a far-forward requirement — a dedicated tagging optics or the second interaction region's secondary focus — and for a tag with a 0.20 GeV near-beam envelope one year yields 1.7×10^7 events and a 5σ sensitivity to a 0.9% modulation.

1 · Introduction

A spin-1 target can absorb two units of photon helicity in the forward Compton amplitude. The structure function that measures this, $\Delta(x, Q^2)$, receives no leading-twist contribution from quarks or from bound spin- $1/2$ nucleons, so in a nucleus it is direct evidence for glue that is not inherited from the individual nucleons — "exotic glue" in the language of Jaffe and Manohar's nuclear gluonometry [1,2]. It has never been measured for any target; the one prior experimental document is a JLab letter of intent for a ${}^{14}\text{N}$ target, without a sensitivity projection [3]. The only absolute normalizations available are moment estimates: a bag-model calculation for the spin- $3/2$ Δ^{++} baryon [4], maximal-transversity scenarios for proton–deuteron Drell–Yan [5], and lattice-QCD matrix elements for the ϕ meson [6] and the deuteron [7]. The EPIOS white paper on polarized ion beams at the EIC names spin-1 ${}^6\text{Li}$ among the nuclei where a discovery is most likely [8], and the companion physics case [9] argues why ${}^6\text{Li}$ — a polarized deuteron bound to an inert α particle, with $A/Z = 2$ and deuteron-like spin dynamics in the ring — is the practical choice.

This paper presents the first numerical projection of a Δ measurement for any target. Two channels are considered. In inclusive DIS on a transversely tensor-polarized ${}^6\text{Li}$ beam the scattered electron's azimuth about the alignment axis acquires a $\cos 2\phi$ modulation proportional to Δ ; only the electron is detected. In coherent diffraction, $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$, the nucleus stays in its ground state and the modulation of the tagged yield mixes the gluonic term with a nuclear-shape term that the polarized-deuteron calculation of Mäntysaari et al. [10] predicts — and ${}^6\text{Li}$, with a quadrupole moment fifty times smaller than ${}^7\text{Li}$'s, separates the two. Section 2 states the observable, Section 3 the simulation framework, Section 4 the Δ model, Section 5 the inclusive projections at truth and reconstructed level, Section 6 the coherent channel with the far-forward requirement it imposes, Section 7 the systematics and assumptions.

2 · The observable

For unpolarized electrons on a spin-1 target whose spin axis makes angle θ_m with the beam, the inclusive cross section per spin projection m is [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1)$$

with ϕ the azimuth of the lepton plane about the virtual photon, measured from the transverse projection of the spin axis. Averaged over a fill, c_m becomes the tensor polarization $P_{zz} = p_+ + p_- - 2p_0$. The b_1, b_2 terms are written in the programme's transcription; in the Hoodbhoy–Jaffe–Manohar convention as applied by HERMES and by Cosyn et al. [11] they carry the opposite sign, which affects only the sign of A_{zz} relative to b_1 and the constant term of the spin-state ratio of §3.3, nothing in the Δ sector. At leading order Δ is generated by the gluon transversity distribution δG through the $O(\alpha_s)$ photon–gluon convolution

$$\Delta(x, Q^2) = -\frac{\alpha_s(Q^2)}{2\pi} \left(\sum_q e_q^2 \right) x^2 \int_x^1 \frac{dy}{y^3} \delta G(y, Q^2)$$

in the sign convention of Ref. [6]. For a transverse spin axis ($\theta_m = 90^\circ$) the measured amplitude and its inversion to the structure function are

$$A \cos 2\phi = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{\Delta} = -\hat{A} \frac{y^2 D_\phi}{1-y}$$

Three properties of this relation shape the analysis. First, at the low y of the sensitive region, $F_2 = 2x(1+R)F_1$ reduces it to $\Delta/F_1 = -2(1+R)\hat{A}$: F_2 cancels exactly (to 2×10^{-16} in the simulation, checked with two independent levers), and $R = \sigma_L/\sigma_T$ is the one structure-function input that reaches the physics result. Second, the amplitude is exactly linear in Δ , so the detector response acts on it as a linear operator, which §5.3 exploits. Third, the target-mass corrections are small: at the four bins of §3.4, $y^2 = 4M^2 x^2/Q^2 \leq 0.006$, so the approximation set of the simulation — $A_{||} = D \cdot g_1/F_1$ with the y^2 -suppressed term dropped, $g_2 = g_2^{WW}$, no target-mass terms — is adequate. One finite- y effect is physical rather than a labelling matter: the tensor-polarized b_1 – b_3 terms feed the $\cos 2\phi$ amplitude at $O(y^2)$, bounded by $1.15 y^2 b_1/F_1$ (the full Cosyn et al. Eq. 17e combination, $\approx 6.9 \times$ the $y^2 b_1/6$ of the b_1 term alone) [11]; for a HERMES-like b_1 this is $\leq 1\%$ of the projected amplitudes at the four bins and is computable from the same data's A_{zz} .

The size of Δ is unknown to within an order of magnitude. The bag-model moment of Sather and Schmidt, $\int x \Delta dx = -0.012 \alpha_s$ with its Q^2 evolution but no x dependence [4], puts the asymmetry at the per-mille level; the explicit $\alpha_s(Q^2)$ is the leading-order Wilson coefficient of the photon–gluon box, so the $O(\alpha_s)$ suppression relative to F_1 is model-independent while -0.012 is the bag estimate. Lattice QCD adds two results, both at unphysical pion masses: the bare ϕ -meson matrix element $A_2 = 0.23(2)(5)$ at $m_\pi = 450$ MeV [6], which converts to $\int x \Delta|_\phi \approx -0.006 \alpha_s$ (our conversion, unrenormalized), and the NPLQCD deuteron δG moment of $-0.010(3)$, bare at $m_\pi \approx 806$ MeV [7] — a transversity-to-momentum ratio roughly ten times below the ϕ 's, implying the opposite sign for the deuteron's Δ moment. Percent-level asymmetries arise only in the maximal-transversity ($\Delta_T g = \Delta g$) Drell–Yan scenarios their authors call optimistic [5]. Nothing exists for ${}^6\text{Li}$ or any $A > 2$ nucleus beyond a $\Delta\Delta$ -isobar estimate for the deuteron [12], so the literature brackets the nuclear Δ in both directions: suppressed, as in the NPLQCD deuteron, or enhanced by binding in analogy with the EMC effect [8]. The projections therefore carry the moment-constrained model of §4 together with the scenario band $\Delta/F_1 \in 10^{-3}$ – 10^{-2} .

3 · Simulation framework

3.1 · Beams, polarization and luminosity

EIC ions are y -matched to the proton rather than rigidity-scaled, because the two rings must share a revolution period, so the ${}^6\text{Li}$ menu is 40.8 GeV/u and 99.5–137.5 GeV/u, the top value being the rigidity cap $275 \text{ GeV} \times Z/A$ [8,13]. The

projections are made at the mid configuration, $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$, $\sqrt{s_{\text{NN}}} = 63 \text{ GeV}$, with the low (5×40.8) and top (18×137.5) configurations used where the hadronic reconstruction or the far-forward acceptance depends on the energy. The in-ring tensor polarization is a placeholder $P_{zz} = 0.6$ (the ANL source targets $P_{zz} \geq 0.8$ [9]; transport to the ring is an open accelerator question, and every statistical error scales as $1/P_{zz}$), the alignment axis is the vertical — the stable spin direction of the arcs, native at the interaction point — and the electrons are unpolarized. Luminosity is quoted per nucleon: $10 \text{ fb}^{-1}/\text{u}$, "one EIC year" in the Yellow Report's accounting (30 weeks at $10^{33} \text{ cm}^{-2} \text{ s}^{-1}$ with 60% efficiency [13]), and $100 \text{ fb}^{-1}/\text{u}$ for the ten-year programme, all of it in the transverse-tensor fill. No lithium luminosity has been published; the yields scale linearly with whatever the machine delivers.

3.2 • Generator and structure functions

No public generator produces polarized e+A events for $A > 1$ — BeAGLE is unpolarized, DJANGO and PEPSI are nucleon-level and longitudinal-only — so the programme's generator, `polligen`, implements only the missing component, the spin-density matrix $\otimes$ structure-function master formula with spin-correlated sampling, and takes the rest from existing tools: the hadronic final state from a PYTHIA 8 production of 8 M events over the three beam configurations. Its validation gates are bin-wise identity with the analytic asymmetry formulas at relative tolerance 10^{-12} , pseudo-experiment spreads that close on every analytic error map, and unbiased $\cos 2\varphi$ recovery under φ -acceptance holes. Because drawing each φ -bin count from a Poisson distribution with the exact expected yield is statistically identical to event-level sampling, a full-luminosity pseudo-experiment costs 24 Poisson draws; event-level sampling is what the closure tests use.

Structure functions are toy parameterizations with PDF-grid backends (CT18NLO, EPPS21, NNPDFpol11) behind one interface, so rates hold to a factor ≈ 1.5 (the toy F_2 is within $\pm 37\%$ of CT18 over the acceptance) and asymmetry shapes are scenarios by construction. $R = \sigma_L/\sigma_T$ is a placeholder of the right magnitude, 0.14–0.18 at the four bins of §3.4, in every projection below; the published SLAC/E143 R1998 world fit [14], all three of its functional forms, sits behind the same hook, and switching to it moves Δ/F_1 by +18, +19, +8 and -4% at the four bins, with a further 0.5–6% from the spread of the fit forms — a correction to apply rather than a band, and the systematic on every Δ quoted below until it is.

3.3 • Estimators

The figures use a single spin state: a binned least-squares fit of $N(\varphi') = C(1 + A \cos 2\varphi')$ rather than the $2\langle \cos 2\varphi' \rangle$ moment, which realistic φ holes bias by more than the signal while the fit stays unbiased, with $\delta A = \sqrt{(2/N)/P_{zz}}$ validated by pseudo-experiments and no $\cos \varphi'$ term, since the m-symmetric tensor fills set it to zero by construction. The azimuth φ' is the covariant one built from the four-vectors [15], equal to $\varphi_e - \varphi_S$ for a massless target and within 5 mrad of it for ${}^6\text{Li}$, and the 25 mrad crossing angle changes only odd harmonics of the electron azimuth because the ePIC axis is the electron beam.

The measurement itself cannot use a single state: the fit then also measures the detector's own $\cos 2\varphi'$ acceptance harmonic divided by P_{zz} , and a 3% harmonic aligned with the vertical spin axis fakes four times the signal. The estimator is therefore the spin-state-sorted ratio of $m = \pm 1$ -rich ($P_{zz} = +0.6$) and $m = 0$ -rich (-1.2) bunches. It cancels any acceptance common to both states bin by bin, turns relative-luminosity errors into a constant, and gives $\delta \hat{A} = 2\sqrt{(2/N)/(P_+ - P_0)} = 0.67 \times$ the single-state error at equal source purity — provided the states alternate bunch by bunch. What the ratio cannot cancel is a difference of acceptance between the states: for two fills the bias is $(\epsilon_2^+ - \epsilon_2^0)/(P_+ - P_0)$ at any luminosity split, so a 10^{-3} difference of the $\cos 2\varphi'$ harmonic fakes 5.6×10^{-4} , 6–13% of the projected amplitudes and 2–6 one-year statistical errors. Bunch-by-bunch alternation, or 10^{-4} stability of the harmonic between fills, is thus a requirement of the measurement rather than a refinement.

3.4 • Acceptance and binning

Rates use per-nucleon luminosity, F_2^A/A , and scattered-electron cuts $|\eta| \leq 3.5$, $E' \geq 0.5 \text{ GeV}$, $0.01 \leq y \leq 0.95$ and $W^2 \geq 10 \text{ GeV}^2$, which accept 5.2×10^9 events in one year (Figure 1). The four displayed (x, Q^2) super-bins are selected by statistical significance within Q^2 bands — an unconstrained ranking concentrates at the lowest Q^2 , whereas the Q^2 lever arm provides the consistency check against power-suppressed contributions — and each merges 3×3 cells of the 40×30 analysis grid. They sit at $(x, Q^2) = (0.028, 1.14)$, $(0.011, 1.14)$, $(0.071, 3.13)$ and $(0.141, 14.3 \text{ GeV}^2)$, i.e. at $y = 0.010, 0.026, 0.011$ and

0.026: the amplitude grows as $(1 - y)/y^2$, so the sensitivity lives at the lowest y the detector can reconstruct, which is what makes the kinematic reconstruction of §5.2 the central detector question.

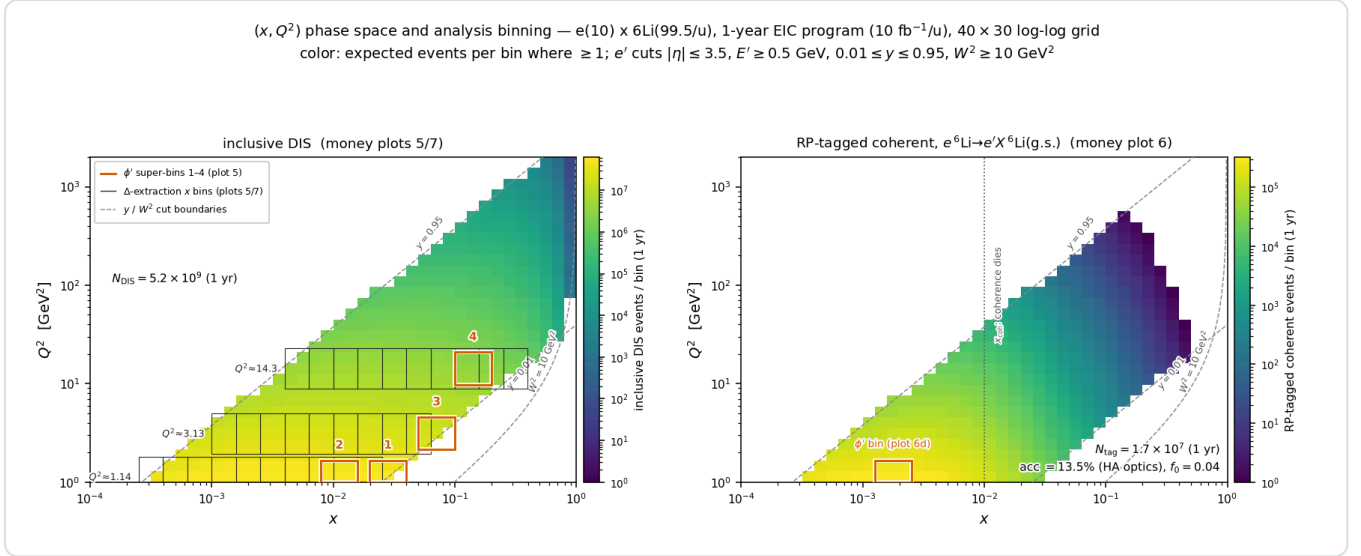


Figure 1 — phase-space coverage and analysis binning at $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$, one EIC year, 40×30 log-log grid; colour: expected events per bin where ≥ 1 ; dashed: y and W^2 cut boundaries. **Left, inclusive DIS:** boxes 1–4 are the ϕ' super-bins of Figure 2; grey combs are the three Q^2 slices ($Q^2 \times [1/1.6, 1.6]$) whose merged x -bin pairs give the data points of the amplitude-versus- x panel and of the Δ extraction (Figures 2 and 3), retained where $N \geq 10^3$ and $\delta A \leq 8 \times 10^{-3}$. **Right, tagged coherent channel:** the same map times $f_{\text{coh}}(x)$ and a 13.5% tag acceptance (§6.1), $N_{\text{tag}} = 1.7 \times 10^7$ in one year, almost entirely at $x < 0.03$; the box is the super-bin of Figure 4(d).

4 • The Δ model

All Δ models reside in one registry (`polli_fastsim/delta_models.py`). The production default, `moment_A`, is an ansatz normalized to the bag moment applied to the per-nucleon Δ of ${}^6\text{Li}$:

$$\Delta_A(x, Q^2) = \frac{1}{3} A \alpha_s(Q^2) F_1(x, Q^2) \frac{x^a(1-x)^b}{N_{\text{peak}}}, \quad \int_0^1 x \Delta(x, \langle Q^2 \rangle) dx = -0.012 \alpha_s \Rightarrow A = -0.292$$

with $(a, b) = (0.7, 3)$. The factor $1/3$ restricts the exotic glue to the polarized deuteron-like pair while the whole-nucleus P_{zz} remains a separate factor, so nothing is double counted; the cluster picture would give ≈ 0.81 instead, an upside of $\times 2.4$ that is not taken. The constraint is imposed at the rate-weighted $\langle Q^2 \rangle = 4.47 \text{ GeV}^2$ of the accepted phase space, and because F_1 evolves the realized moment varies by a few per cent across the plotted Q^2 slices; α_s comes from the CT18NLO table.

`moment_B` imposes the same constraint without the F_1 factor and yields amplitudes a factor 2–8 lower; the A-versus-B spread is the dominant ansatz systematic, and both curves appear on every plot. Porting a single-hadron bag moment to a per-nucleon nuclear structure function has no precedent, so normalization and x dependence are both scenario choices, bracketed by `moment_B` from below and possible nuclear enhancement from above.

5 • Inclusive projections

5.1 • Truth level

Figure 2 shows the ϕ' pseudo-data in the four super-bins with one-year statistics and the extracted amplitude versus x along the $Q^2 = 1.14 \text{ GeV}^2$ slice. The injected amplitudes are $7.4, 4.4, 9.5$ and 9.5×10^{-3} against one-year errors of $1.7, 1.4, 2.8$ and 4.5×10^{-4} — $44, 31, 34$ and 21σ — and ten-year errors of $0.55, 0.45, 0.87$ and 1.4×10^{-4} ; the fitted values agree with the injected ones within the errors. Under the moment constraint the measurement is not a null test: the `moment_A`-versus-`moment_B` spread is resolved by the x dependence of the amplitude, and even a flat $\Delta/F_1 = 10^{-3}$ scenario remains a $\geq 3\sigma$ effect per bin at $P_{zz} = 0.3$. At ten-year precision the statistical errors sit below the polarimetry scale systematic (3% of an amplitude of 10^{-2} is 3×10^{-4}), so the ten-year measurement is systematics-limited.

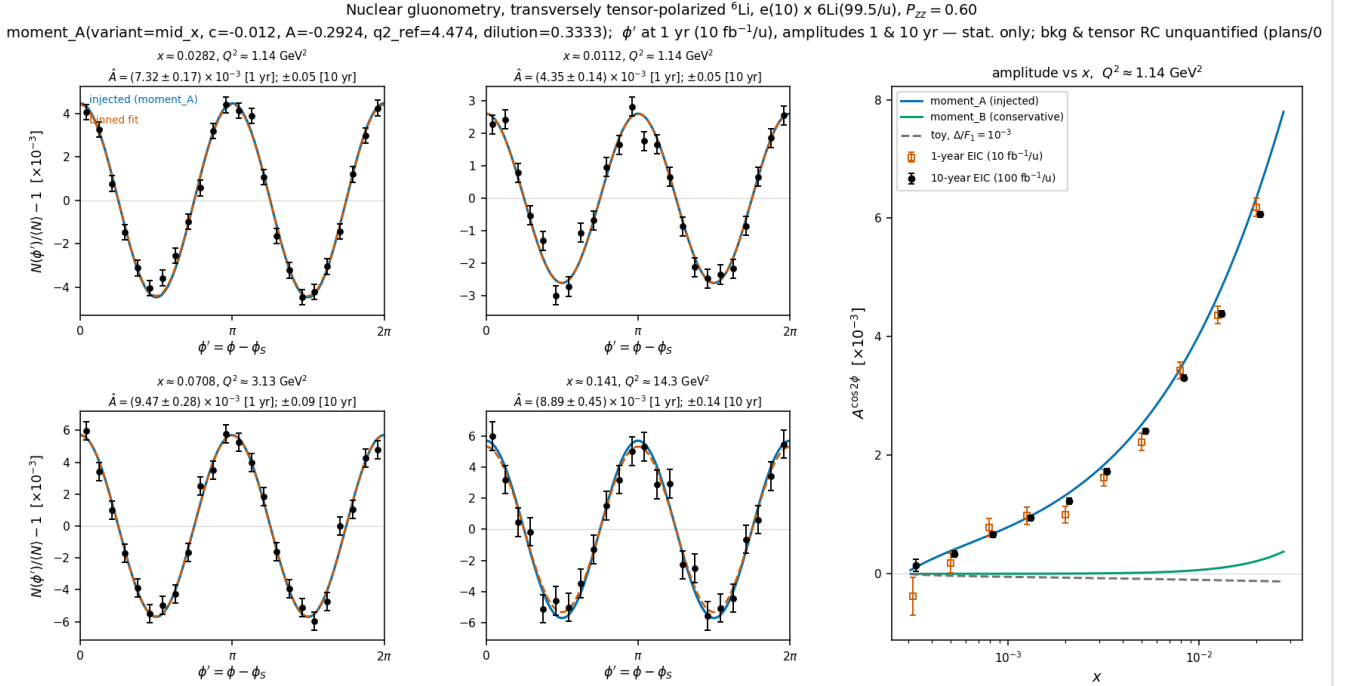


Figure 2 — inclusive gluonometry at $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV}/u)$, $P_{ZZ} = 0.60$, `moment_A` with dilution 1/3. Left 2x2: ϕ' pseudo-data with one-year statistical errors in the four super-bins ($Q^2 = 1.1 \rightarrow 14 \text{ GeV}^2$, $x = 0.011\text{--}0.14$); injected curve (blue), binned fit (vermillion); annotations give δA for both programmes. Right: extracted amplitude versus x with one-year (open) and ten-year (filled) errors against the `moment_A`, `moment_B` and flat- 10^{-3} curves. Only the scattered electron is detected.

Figure 3 unfolds the same pseudo-measurements to the structure function, $x\Delta(x, Q^2)$ per nucleon at the three sweet-spot Q^2 slices, with independent one- and ten-year draws and the model bin-centering factor $K = \Delta_{\text{model}}(x_c, Q^2_c)/A_{\text{model, bin}}$; the area under each curve is the ported bag moment ($\div 3$ dilution). The best bins reach $\Delta = -0.135 \pm 0.003$ at $(x, Q^2) = (0.02, 1.14)$, -0.070 ± 0.002 at $(0.05, 3.13)$ and -0.0047 ± 0.0004 at $(0.32, 14.3)$ in one year, i.e. 2–9% relative, and three times better in ten.

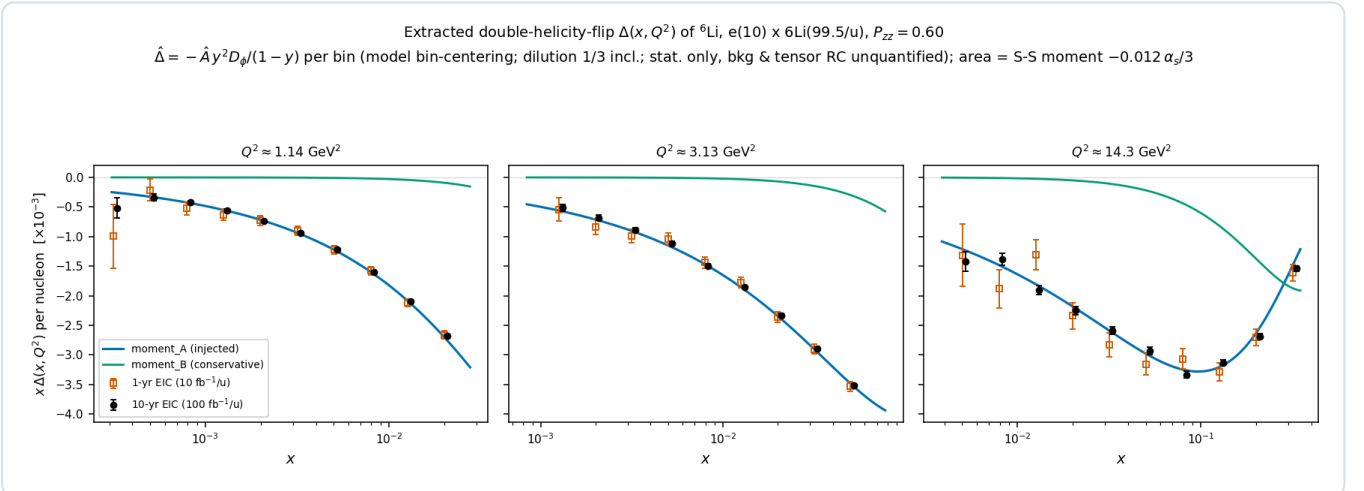


Figure 3 — the extracted $\Delta(x, Q^2)$: the amplitudes of Figure 2 unfolded to $x\Delta$ per nucleon at the three sweet-spot Q^2 slices (model bin-centering, $\delta\Delta = \delta A \cdot |\partial\Delta/\partial A|$), with independent one-year (open) and ten-year (filled) pseudo-measurements against the `moment_A` and `moment_B` curves.

5.2 • Reconstructed level

Three of the four super-bins sit at $y \approx 0.01$, where the electron method alone gives $\delta y/y = 0.5\text{--}1.2$ and the bin is not reconstructible; x must come from a hadronic-method y . The reconstructed-level pseudo-experiments of the companion note [16] pass every event through the lab frame, EMCal energy and track-angle smearing, the head-on transformation, the electron-method Q^2 , a hadronic y , the mixed-method x , the analysis cuts on reconstructed quantities, an η -dependent electron identification efficiency and the covariant ϕ' , then form the spin-state ratio of the two fills with a $3\% \cos 2\phi'$ and

2% $\cos \varphi'$ acceptance harmonic and a 10^{-3} relative-luminosity offset switched on. Table 1 gives the result with the documented 25% Gaussian stand-in for the hadronic y — the ePIC kinematic-fit study's smearing and the ATHENA value at $y \approx 0.01$ [17] — and with the y of the PYTHIA 8 final state through the hadron-side detector response (Σ method, 50 MeV of calorimeter noise).

SUPER-BIN	25% HADRONIC-Y STAND-IN			PYTHIA 8 FINAL STATE, Σ METHOD			$\Delta\hat{A}$ (1 YR)	$\Delta\hat{A}$ (10 YR)
(x, Q^2)	PURITY	EFFICIENCY	D	$\Delta y/y$	PURITY	D	[10^{-4}]	[10^{-5}]
0.028, 1.14	0.66	0.42	0.92	0.32	0.43	0.79	1.2	3.7
0.011, 1.14	0.63	0.59	0.99	0.22	0.54	0.85	0.9	2.9
0.071, 3.13	0.69	0.38	0.90	0.29	0.47	0.82	1.6	5.2
0.141, 14.3	0.69	0.66	0.96	0.15	0.69	0.95	3.0	9.3

Table 1 — the four super-bins at the reconstructed level (mid configuration, spin-state ratio of the $+0.6 / -1.2$ fills). Purity: fraction of the reconstructed bin's rate that originates in the true bin; efficiency: fraction of the true-bin rate reconstructed in the bin after electron identification; D: rate-weighted, φ' -diluted amplitude of the events in the reconstructed bin over that of the true bin; $\delta y/y$: 68% half-width of the Σ -method resolution at the spot. The statistical errors are the same for both hadronic- y models to within 3%.

Three conclusions follow. The fit is unbiased at the reconstructed level: every $\hat{A}$ agrees with its reconstructed-bin truth within about two standard errors while the acceptance harmonic and the luminosity offset are on, and the $\sin 2\varphi'$ coefficients are consistent with zero. The errors are 0.59–0.69 of the single-state truth-level values, because the $1.5\times$ gain of the $m = 0$ -rich fill outweighs the reconstruction efficiency of 0.38–0.66 — presuming the source delivers $m = 0$ -rich bunches at $|P_{zz}| = 1.2$ with the same purity as the ± 1 fill at 0.6. And the real final state costs purity rather than statistics: PYTHIA puts 0.11 of the hadronic $E - p_z$ sum into neutral hadrons, which belong to the HCal, so the purities fall to 0.43–0.69 and the reconstructed amplitude sits 7–23% below the true-bin value rather than 1–10%, and more of the answer passes through the bin-centering factor. The resolution degrades with beam energy because the sweet spots move to lower y as s grows, and the noise floor decides it below $y \approx 0.02$ — the Σ -method $\delta y/y$ at $y = 0.005$ runs $0.22 \rightarrow 0.33 \rightarrow 0.54 \rightarrow 1.01$ for $0 \rightarrow 25 \rightarrow 50 \rightarrow 100$ MeV of calorimeter noise on a hadronic $E - p_z$ sum of 0.2–0.5 GeV, a regime no published ePIC study covers. The $x \approx 0.14$ bins are best measured at the low-energy configuration, where their $\delta y/y$ is 0.24 and 0.11. Unfolded to Δ , the best reconstructed bins reach $\delta\Delta = 2.4\times 10^{-3}$ at $Q^2 = 1.14 \text{ GeV}^2$ and 1.2×10^{-3} at 3.13 GeV^2 in one year, with purities ≈ 0.55 .

5.3 • Bin centering

Correcting the reconstructed amplitude to a point value of Δ with the model's own K is circular. Because the amplitude is exactly linear in Δ , the response can instead be applied as a linear operator and a $\Delta(x)$ shape fitted through it per Q^2 slice on a bounded grid. Correcting the same pseudo-data with either of two deliberately wrong priors, the model dependence of K at the four super-bins falls from $(-4.1, +8.0, -5.5, +4.9)\%$ to $(-1.5, -1.8, -0.3, +0.5)\%$ with the `moment_B` prior and from $(+8.6, +2.5, +7.3, +3.2)\%$ to $(+4.6, +3.0, +1.8, -0.9)\%$ with the flat toy, and over the plotted bins of Figure 3 from up to 29% to $\leq 6\%$. The Δ error bar then carries four terms — statistics, the shape fit, the response Monte Carlo (0.1–1% per bin) and the spread over priors the shape family cannot absorb — of which the last dominates.

6 • The coherent channel

6.1 • Detecting the intact recoil

In coherent diffraction the recoil retains the beam's magnetic rigidity — $A/Z = 2$ exactly, and the diffractive momentum loss $x_p < 0.01$ changes it by less than 1% — and emerges at $\theta = p_T/p \lesssim 1$ mrad, inside the beam. Of the far-forward system at IP6 — the B0 spectrometer, the off-momentum detectors, the Roman Pots and the zero-degree calorimeter, with the acceptance windows of Refs. [18–20] — only the pots can see such a recoil, and only where it clears both the 10σ beam envelope and the pots' own aperture. Both are constraints on the *angle* at the interaction point, so with an exponential t distribution the tagged fraction is

$$\varepsilon_{\text{tag}} = \exp(-B p_{T,\text{cut}}^2), \quad p_{T,\text{cut}} = \theta_{\text{cut}} A p_u, \quad \theta_{\text{cut}} = \max(10 \sigma_\theta, \theta_{\text{aperture}})$$

an exponential in the square of the beam divergence times the beam momentum. Table 2 evaluates it with $B = 50 \text{ GeV}^{-2}$ at the three ${}^6\text{Li}$ configurations, using the Yellow Report's divergences per configuration and optics — a γ -matched ion has the proton's $\beta\gamma$ and hence the proton's divergence at equal normalized emittance, and pays a factor $\sqrt{2}$ only where it is rigidity-capped, i.e. at the top configuration [13] — and the pots' aperture for an intact ${}^6\text{Li}$, measured by tracking one through the ePIC far-forward geometry: the boundary is $|\theta_x| \gtrsim 2.0 / 1.35 / 1.03 \text{ mrad}$ in the $5 \times 41 / 10 \times 100 / 18 \times 275$ optics against $|\theta_y| \gtrsim 1.8\text{--}3 \text{ mrad}$, because the transport images an interaction-point angle onto the pot plane with a horizontal lever $R_{12} \approx 30.6 \text{ m}$ against a vertical few metres [16].

${}^6\text{Li}$ CONFIGURATION	P_{ION}	Σ_θ H/V, HIGH ACCEPTANCE [MRAD]	$E_{\text{TAG}}, 10\Sigma$ ENVELOPE	POT APERTURE $\theta_x $	$E_{\text{TAG}},$ APERTURE	TAGGING OPTICS: Σ_θ, B^* FACTOR
5×40.8	245 GeV	220 / 380	5×10^{-7}	2.0 mrad	1×10^{-6}	58 μrad , $\times 14.5$
10×99.5	597 GeV	180 / 180	8×10^{-26}	1.35 mrad	8×10^{-16}	24 μrad , $\times 58$
18×137.5	825 GeV	92 / 92	4×10^{-13}	1.03 mrad	2×10^{-17}	17 μrad , $\times 29$

Table 2 — the coherent tag at IP6 with published inputs. Divergences: Yellow Report Table 10.1 with the γ -matched species step [13]; at 41 GeV the high-acceptance and high-divergence optics are identical, the only configuration where the Yellow Report offers no separate high-acceptance option. Aperture: the September-2024 `epic-main` geometry, one event per scan point, no beam envelope; the pot layout has since changed (16 mm modules, per-energy insertions) and must be re-measured. Tagging optics: the divergence that puts the 10σ envelope at $p_T = 1/\sqrt{B} = 0.14 \text{ GeV}$, where $L \times \varepsilon_{\text{tag}}$ is maximal and $\varepsilon_{\text{tag}} = 1/e$, and the β^* increase over the high-acceptance optics it implies.

The conclusion is unambiguous: with any published optics the coherent intact- ${}^6\text{Li}$ channel does not exist at IP6, at any energy. It exists under two conditions. A *tagging optics*, with β^* raised until the envelope sits at $p_T = 1/\sqrt{B}$, where luminosity times acceptance is maximal, gives $\varepsilon_{\text{tag}} = 1/e$ at every configuration; the smallest ask is at 5×41 , β^* from 0.9 to 13 m, well inside the forward-physics optics demonstrated at the LHC — but the pots must then approach to 0.58 mrad where the silicon now sits at 2 mrad, so the two levers are multiplicative and the second is the near-beam question of Report 4 [21]. Alternatively, the *secondary focus* of the second interaction region ($\sigma \approx 0.15 \times 0.1 \text{ mm}$ at $\sim 45 \text{ m}$) lets pots at $z = 44.0/45.5 \text{ m}$ reach $p_T \approx 0$, with published intact-recoil efficiencies of 47% (d), 32% (${}^3\text{He}$), 29% (${}^4\text{He}$) and 17.8% (${}^7\text{Li}$) at top energy, rising steeply at lower energies [22]; no ${}^6\text{Li}$ case is published and our interpolation is $\approx 20\%$. Either option delivers a tag of order 10–40%, and the projections below are made for a near-beam envelope of 0.20 GeV — $\varepsilon_{\text{tag}} = 13.5\%$ (9–20% for $B \in 40\text{--}60 \text{ GeV}^{-2}$) — as the physics case for providing it. Even so, one degeneracy remains: a Roman-Pot hit measures A/Z , and an intact ${}^6\text{Li}$, an α and a d at the same velocity share trajectory and timing. The second-detector design answers it with a Z^2 -scaling Cherenkov counter behind the pots [19]; for IP6 no charge-discrimination study exists, though the AC-LGAD readout digitizes an 8-bit charge over four planes and the $\alpha + d$ pair separates by hit topology [21] (open item #19 of the programme).

6.2 • Rate model

$$\sigma_{\text{coh}} = f_{\text{coh}}(x) \sigma_{\text{DIS}}, \quad f_{\text{coh}}(x) = \frac{f_0}{1 + (x/x_{\text{coh}})^2}, \quad f_0 = 0.04 [0.02\text{--}0.08]$$

No diffractive prediction exists for a light nucleus (the lightest published case is Ca), so the f_0 band brackets HERA ep diffraction (10–15% of DIS [23]) and heavy-nucleus saturation estimates (20–25% [24]); parameterizing f_{coh} in Bjorken x rather than x_p is optimistic for high-mass diffraction, which the band is meant to cover. The scale $x_{\text{coh}} = 0.01$ marks where the coherence length $l_c = 0.105 \text{ fm}/x_p$ falls below the ${}^6\text{Li}$ size, and the slope $B = 50 \text{ GeV}^{-2}$ (band 40–60) follows from $B = \langle r^2 \rangle / 3$ with the matter radius (the charge radius gives ≈ 57 [25]; gluonic distributions may be wider [26]), valid below the form-factor minimum at $|t| \approx 0.31 \text{ GeV}^2$ [27,28]. One year produces 1.2×10^8 coherent events, of which the 0.20 GeV tag

keeps 1.7×10^7 (1.7×10^8 in ten years); the best super-bin, $x \in [0.0013, 0.0025]$, $Q^2 \in [1, 1.7]$ GeV², holds 1.75×10^6 and gives $\delta A = 1.8 \times 10^{-3}$ (one year) and 0.6×10^{-3} (ten years), i.e. 5σ floors at modulations of 0.9% and 0.3%.

6.3 • The modulation and its two mechanisms

The $\cos 2\phi'$ modulation of the tagged yield carries two mechanisms. The *deformation* term is scaled from the polarized-deuteron colour-glass calculation of Mäntysaari et al. [10], the only published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all forward citations checked; the group's subsequent work is unpolarized [29,30]). Their coefficients show a_2 growing linearly in $|t|$ with $a_2(m=0) = -2a_2(m=\pm 1)$, from the 6% D-wave modulating the diffractive slope. For a fill with populations p_m the ensemble coefficient is proportional to P_{zz} :

$$a_2(t; P_{zz}) = -\frac{P_{zz}}{4} \varepsilon_{B0} B |t|, \quad \langle a_2 \rangle_{\text{tag}} = a_2(p_{T,\text{cut}}^2 + 1/B) = 0.036 \quad [0.018-0.059]$$

Two conventions matter. The anchor's Eq. (9) expands $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$ with Φ the azimuth of the recoil relative to the polarization axis in the γ^*d frame, so the $\cos 2\Phi$ coefficient is $2a_2$: the $\langle a_2 \rangle$ quoted here and injected in Figure 4(d) understates the deformation modulation by a factor two (0.072 rather than 0.036 at $P_{zz} = 0.6$), and the coherent numbers of this paper are conservative by $\times 2$. And the modulation is one of the *recoil* azimuth $\phi_t - \phi_S$, not of the electron azimuth used as the axis of Figure 4(d); the reconstructed-level analysis fits both azimuths [16].

$\varepsilon_{B0} \equiv \Delta B_0/B$ is scaled from the deuteron value of 0.21 by two indicators that disagree: the relative quadrupole deformation $Q/\langle r^2 \rangle$ of ${}^6\text{Li}$ (0.012) is about one fifth of the deuteron's (0.063) and opposite in sign, while the α -d asymptotic D/S ratio is disputed — a transfer measurement gives $\eta = +0.0003(9)$, consistent with zero [31], against older sub-Coulomb analyses favouring $\eta \approx -0.01$. We adopt $\varepsilon_{B0} \in (-0.04-0.13)$ with default -0.08 , leaving near-zero deformation open; the 5σ floors cover its detection or exclusion. The band's width is genuine theoretical uncertainty: even exact few-body calculations miss the ${}^6\text{Li}$ quadrupole cancellation by a factor ≈ 3 [32], and the gluonic deformation need not share the cancellation of the charge sector. The sign flip relative to the deuteron is a testable prediction ($Q({}^6\text{Li}) < 0$), and tensor modulations of comparable size are established at these $|t|$: the elastic e-d $T_{20} = -(0.1-0.3)$ [33] and HERMES $A_{zz}(d) \approx -1\%$ at $x \approx 0.01$ [34], the latter attributed to coherent double scattering [35–37]. The flat gluon-transversity term is separately bounded at $3 \times 10^{-3} - 10^{-2}$ (at $P_{zz} = 1$) from the bare lattice ϕ transversity [6] divided by the $\sim 10\times$ NPLQCD deuteron suppression [7], with the maximal Drell–Yan scenario [5] as the ceiling. The equal-rate population average behind $\langle a_2 \rangle$ carries a one-sided $\times 0.73$ rate-weighting systematic, inside the ε_{B0} band.

The two mechanisms separate by shape: deformation is linear in t , sign-flipped relative to the deuteron and expected to weaken toward small x_p (JIMWLK evolution washes it out [30]), while exotic glue is flat in t and persists at small x , where it connects to the elliptic-gluon programme [38]. In a two-component fit $a_2(t) = c_{\text{def}} \cdot |t| + a_g$ per x_p bin, both components exceed the statistical floor by $\geq 5\sigma$ at $100 \text{ fb}^{-1}/u$ for the 0.20 GeV tag. Incoherent breakup is m-state-blind in this mechanism [10]: it dilutes the modulation but cannot mimic it. And deformation-driven modulations, the polarized-deuteron imaging signal, are bounded on ${}^6\text{Li}$ by its quadrupole moment, $Q = -0.0806(6) \text{ fm}^2 \approx Q({}^7\text{Li})/50$ [39]: a sizable flat-in- t $\cos 2\phi$ on ${}^6\text{Li}$ cannot be a shape effect, and the mechanisms entangled for the deuteron separate in the ${}^6\text{Li}/d$ comparison — the null test unavailable to the deuteron.

Coherent $e^6\text{Li} \rightarrow e'X^6\text{Li}(\text{g.s.})$, transversely tensor-polarized, $e(10) \times 6\text{Li}(99.5/u)$, $P_{zz} = 0.60$
 nario rates; modulation anchored on the polarized- d CGC calculation, PLB 858:139053 -- plans/06 SS6.4b; statistical only, backgrounds/purity and tensor RC unquanti

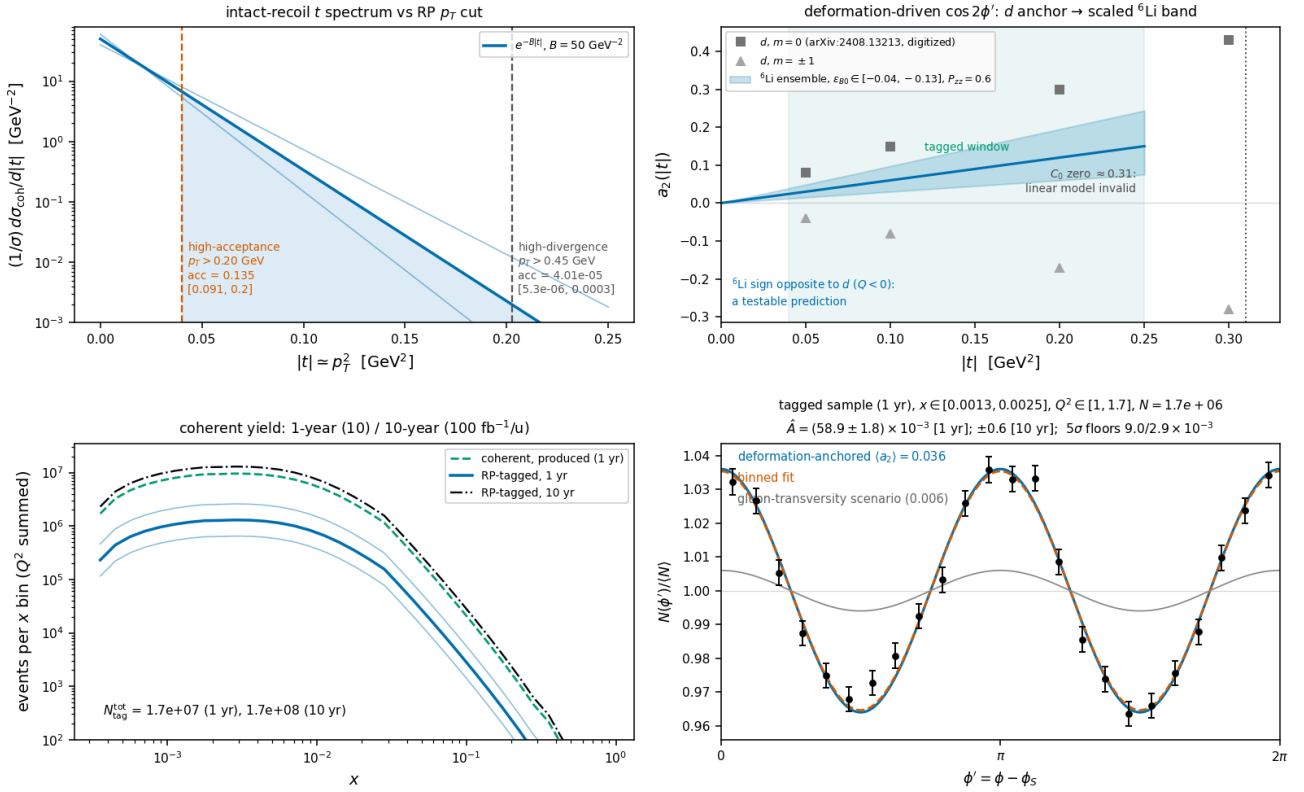


Figure 4 — the coherent channel $e^6\text{Li} \rightarrow e'X^6\text{Li}(\text{g.s.})$ at $e(10 \text{ GeV}) \times ^6\text{Li}(99.5 \text{ GeV}/u)$, $P_{zz} = 0.60$, for a 0.20 GeV near-beam envelope. (a) Recoil t spectrum with the slope band and the 0.20 and 0.45 GeV cuts (13.5% and 4×10^{-5}). (b) Digitized deuteron a_2 per m state [10] against the scaled ^6Li ensemble band (sign flip predicted), with the tagged window and the form-factor-zero exclusion. (c) Tagged yield versus x for both programmes with the f_0 band. (d) ϕ' pseudo-data (one-year statistics) at the anchored $(a_2) = 0.036$ with the flat gluon-transversity scenario (grey); extraction $\hat{A} \pm 1.8 \times 10^{-3}$ (one year), $\pm 0.6 \times 10^{-3}$ (ten years).

That the analysis closes once a tag exists has been demonstrated with the reconstructed-level chain [16]: the recoil is transported to the pots with divergence smearing and an aperture, the spin-state ratio is formed in the two azimuths and fitted with an acceptance-weighted template basis in which the pot geometry's own $\langle \cos 2\beta \rangle$ cancels, and reflection-forbidden sin terms serve as null columns that resolve a pot roll to 5–8 mrad. For a 0.73 mrad approach at the 5×41 configuration — acceptance 7.8%, 1.4×10^6 tagged events in the lowest of four $|t|$ bins in one year — the fit recovers the deformation coefficient $a_t = 0.115 \pm 0.003$, 0.181 ± 0.004 and 0.260 ± 0.009 against 0.119, 0.179 and 0.257 injected in the three bins that carry the statistics, and the flat term to ± 0.0014 in the lowest bin, where two independent pseudo-experiments return 0.0102 and 0.0151 against 0.010 injected — a residual bias of the template basis at this strongly anisotropic aperture, of the order of the one-year error, that a ten-year measurement would have to remove.

6.4 • Backgrounds

BACKGROUND	WHY IT FAKES THE TAG	HANDLE
$\alpha + d$ breakup (threshold 1.474 MeV; the 2.186 MeV 3^+ state, $\Gamma = 24 \text{ keV}$, decays $\sim 100\%$ to $\alpha + d$ [40])	Both fragments at beam rigidity, on the trajectory and velocity of an intact ^6Li : tracking and time-of-flight blind; only dE/dx ($\propto Z^2$: 9/4/1) or the two-hit topology separates	Roman-Pot charge discrimination (open item #19, §6.1). Fallback: two-component $ t $ fit (coherent $e^{-50 t }$ against a $\sim 10\times$ flatter incoherent slope; crossover at 0.05–0.2 GeV 2 , inside the window [26,41]); $e + \text{Pb}$ benchmark 80–99% incoherent rejection [42]
Nucleon-emission breakup: $\alpha + p + n$ (3.70 MeV), $^3\text{He} + t$ (15.79 MeV) [40]	A fragment could pass unnoticed	$p \rightarrow$ off-momentum detectors ($R = 0.50$), $n \rightarrow$ ZDC, $^3\text{He} \rightarrow$ Roman-Pot window ($R = 0.75$): vetoable; inefficiency is second order

${}^6\text{Li}^*$ 3.5629 MeV (0^+ , $T = 1$, $\Gamma = 8.2$ eV) [40]	Particle-stable ($\alpha+d$ parity-forbidden); γ -decays to a real $A = 6$, $Z = 3$ track	Isoscalar exchange cannot drive $T = 0 \rightarrow 1$; the boosted 0.1–0.4 GeV γ lands mostly in the B0 calorimeter (photons ≥ 50 MeV, 5.5–20 mrad [20]), tail in the ZDC; the 5.366 MeV $2^+ T = 1$ state always emits a nucleon. Every $T = 1$ state is γ - or nucleon-vetoable; only $T = 0$ states feed the blind $\alpha+d$ channel
Bethe–Heitler / QED Compton	Intact nucleus by construction	Calculable (t dependence = charge form factor squared [27,28]); subtract and reuse as an acceptance candle; its charge-quadrupole $\cos 2\varphi$ is $< 10^{-3}$ for $ t < 0.15$ GeV ² [28,32]
Beam-gas / halo; e^- -side fakes	Accidental beam-rigidity tracks; π / pair-symmetric e^- at high y	Non-colliding bunches, ~ 30 ps timing, vertex association; standard DIS e^- identification plus (e^- , recoil) kinematic consistency

Table 3 — backgrounds to the coherent tag, ranked; the full budget is in the programme's plan documents. The $\alpha + d$ channel is the one that decides the purity.

7 • Systematics and assumptions

Two extraction systematics survive the spin-state ratio: the polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$, common with A_{zz} and with no established tensor polarimetry method [8], and radiative corrections for tensor observables, which have not been calculated; the unpolarized QED tail is calculable and doubles as calibration. The detector nuisances have been measured rather than argued, holding the response fixed and varying one thing with common random numbers: a $\pm 1\%$ electron energy scale moves Δ by 0.3–0.9% at the four bins, a hadronic-resolution mismatch (0.30 generated, 0.25 corrected) by 0.5–1.4%, an η tilt of the electron-identification efficiency by $\leq 0.03\%$ and the Yellow Report's η -dependent calorimeter table by $\leq 0.1\%$, all above the 0.05–0.34% Monte Carlo floor and below the bin-centering model dependence of §5.3. In interpretation, a $\Delta\Delta$ -isobar admixture can mimic Δ at leading twist [12] and conventional double scattering generates the low- x b_1 [35–37]; the Q^2 lever arm and the ${}^6\text{Li}/d$ comparison discriminate against both. For the coherent channel, an error in the unpolarized $\cos 2(\alpha - \beta)$ harmonic u_2 reaches the flat term at first order through $a_t \delta u_2 \langle \cos 2\beta \rangle$ — a ZEUS- 1σ δu_2 [43] moves a_e by 20% — and a fill-to-fill change of the pot envelope is amplified by the aperture's anisotropy: with the slot-shaped cutout of the earlier study a 10^{-3} change biased a_t by 19%, so the pots' envelope must be stable to 10^{-4} between the spin states, or the states must alternate bunch by bunch. Table 4 lists the assumptions and what a wrong one costs.

#	ASSUMPTION	IMPACT IF WRONG
1	A near-beam tag for the intact recoil exists — a tagging optics with pots that follow, or IR-8 (§6.1); the projections take a 0.20 GeV envelope	No tag, no coherent channel; the inclusive measurement is unaffected. The pot aperture is conditioned on a superseded geometry and must be re-measured
2	Coherent fraction $f_0 = 0.04$ [$\times 2/\div 2$], in Bjorken x (§6.2); no diffractive model for a light nucleus exists	Yields scale linearly; δA as $1/\sqrt{f_0}$
3	Bag moment ported to per-nucleon ${}^6\text{Li}$ with dilution 1/3; x shape ours (§4)	Overall Δ scale; the literature brackets both directions [7,8]; the cluster dilution 0.81 is a $\times 2.4$ upside
4	ε_{B0} scaled from the deuteron by two indicators disagreeing by $\times 3$ (§6.3)	Deformation amplitude anywhere in the band, which every plot carries; the sign flip is robust
5	No event-by-event Z identification in the pots; purity from the $ t $ -shape fit (§6.4)	Z identification would improve purity; an incoherent slope model that is off grows the fit systematic
6	Linear $a_2(t)$ for $ t < 0.2$ GeV ² only; the form-factor zero at 0.31 GeV ² is excluded, not modelled	The upper tagged window carries unmodelled structure
7	Toy structure functions with grid backends and a placeholder R ; no target-mass terms; $g_2 = g_2^{WW}$ (§3.2)	Rates to $\times 1.5$; shapes are scenarios; the published R moves Δ/F_1 by -4 to $+19\%$, its fit-form spread by 0.5–6%
8	All luminosity in transverse-tensor fills at $P_{zz} = 0.6$, the $m = 0$ -rich fill at -1.2 with equal purity, alternating bunch by bunch (§3.3)	N scales with the run-plan share and the amplitude with P_{zz} ; without alternation a 10^{-3} acceptance difference fakes 2–6 one-year errors

9	Impulse approximation; no final-state interactions; tensor radiative corrections unknown	FSI distorts the recoil spectrum at larger $ t $; RC unquantified
10	A calorimeter noise floor of 50 MeV on the hadronic $E - p_z$ sum (§5.2)	The purities of Table 1 degrade in proportion; at 100 MeV the $y \approx 0.01$ bins are lost at the mid configuration

Table 4 — the assumptions behind the projections, ranked by leverage.

8 · Summary and outlook

A transversely tensor-polarized ${}^6\text{Li}$ beam at the EIC measures the double-helicity-flip structure function $\Delta(x, Q^2)$ — never measured for any target — in inclusive DIS with the scattered electron alone, at $21\text{--}44\sigma$ per (x, Q^2) bin in one year under the moment-constrained scenario and at $\geq 3\sigma$ per bin for a flat 10^{-3} amplitude even at $P_{zz} = 0.3$, discriminating between interpretations of the bag moment by the x dependence and extracting $x\Delta(x, Q^2)$ to $2\text{--}9\%$ in the best bins. The result survives reconstruction: with the spin-state ratio of $m = \pm 1$ -rich and $m = 0$ -rich bunches and a hadronic-method y from a PYTHIA 8 final state, the errors are $(0.9\text{--}3.0)\times 10^{-4}$ and the amplitude unbiased, the price being bin purities of $0.43\text{--}0.69$ that the calorimeter noise floor at a hadronic $E - p_z$ of $0.2\text{--}0.5$ GeV controls. The measurement makes three demands on the machine that are specific and answerable: tensor polarization delivered as alternating $m = \pm 1$ -rich and $m = 0$ -rich bunches, transverse at the interaction point; relative luminosity between the spin states controlled to 10^{-4} , or the alternation that makes it a constant; and a tensor polarimetry scale at the few-per-cent level.

The coherent channel carries the same physics with a null test the deuteron cannot offer, and a predicted sign flip of the shape term. It is also a requirement rather than a projection: with the Yellow Report's beam divergences and the measured pot aperture, the intact ${}^6\text{Li}$ cannot be tagged at IP6 with any published optics, at any energy. What recovers it is a tagging optics with the envelope at $p_T = 1/\sqrt{B}$ — a β^* increase of $15\times$ at 5×41 GeV, with pots that approach to 0.6 mrad — or the secondary focus of the second interaction region; either delivers a tag of $10\text{--}40\%$, for which one year yields 1.7×10^7 events and a 5σ sensitivity to a 0.9% modulation, against a deformation-anchored 3.6% and a gluonic term of $0.3\text{--}1\%$. The highest-leverage inputs from outside the programme are, on the machine side, a light-ion optics with β^* chosen for tagging and a charge-identification study of the pot readout; on the theory side, a rerun of the IP-Glasma polarized-deuteron setup with an α -d cluster density, which would replace both ϵ_{B0} indicators and the f_0 band at once, a lattice Δ moment for a light nucleus, and radiative corrections for tensor observables.

References

- [1] R. L. Jaffe, A. Manohar, *Nuclear gluonometry*, Phys. Lett. B 223 (1989) 218.
- [2] P. Hoodbhoy, R. L. Jaffe, A. Manohar, *Novel effects in deep inelastic scattering from spin-one hadrons*, Nucl. Phys. B 312 (1989) 571.
- [3] J. Maxwell et al., *Search for exotic gluonic states in the nucleus*, JLab LOI12-16-006, arXiv:1803.11206.
- [4] E. Sather, C. Schmidt, *Size and scaling of the double-helicity-flip hadronic structure function*, Phys. Rev. D 42 (1990) 1424.
- [5] S. Kumano, Q.-T. Song, Phys. Rev. D 101 (2020) 054011, arXiv:1910.12523; Phys. Rev. D 101 (2020) 094013, arXiv:2003.06623.
- [6] W. Detmold, P. E. Shanahan, Phys. Rev. D 94 (2016) 014507 [Erratum: Phys. Rev. D 95 (2017) 079902], arXiv:1606.04505; W. Detmold, D. Pefkou, P. E. Shanahan, Phys. Rev. D 95 (2017) 114515, arXiv:1703.08220.
- [7] F. Winter et al. (NPLQCD), Phys. Rev. D 96 (2017) 094512, arXiv:1709.00395.
- [8] G. Atoian et al. (EPIOS), *Realizing the scientific program with polarized ion beams at EIC*, Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794.
- [9] C. Peng, *Deep inelastic scattering on polarized lithium: the physics case for ${}^6\text{Li}$ and ${}^7\text{Li}$ beams at the Electron-Ion Collider*, Report 0 of this series (2026).
- [10] H. Mäntysaari, F. Salazar, B. Schenke, C. Shen, W. Zhao, *Spatial imaging of polarized deuterons at the Electron-Ion Collider*, Phys. Lett. B 858 (2024) 139053, arXiv:2408.13213.
- [11] W. Cosyn, B. Roldan Tomei, A. Sosa, A. Zec, Eur. Phys. J. A 61 (2025) 83, arXiv:2410.12764.
- [12] M. Nzar, P. Hoodbhoy, *Estimation of the double helicity flip deuteron structure function*, Phys. Rev. D 45 (1992) 2264.
- [13] R. Abdul Khalek et al., *EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419 (Tables 10.1–10.2, §10.1); C. Peng, *EIC and ePIC: the machine and the detector this programme is projected on*, Report 3 of this series (2026).
- [14] K. Abe et al. (E143), Phys. Lett. B 452 (1999) 194, hep-ex/9808028.
- [15] A. Bacchetta, M. Diehl, K. Goeke, A. Metz, P. J. Mulders, M. Schlegel, JHEP 02 (2007) 093, hep-ph/0611265.
- [16] C. Peng, *From detector signals to $\Delta(x, Q^2)$: the reconstruction chain of the inclusive and coherent $\cos 2\phi$ measurements*, Report 2 of this series (2026).

[17] S. Maple (ePIC), *Tracking and inclusive DIS reconstruction with the ePIC detector at the EIC*, seminar, Birmingham, 11 December 2024; ATHENA Collaboration, JINST 17 (2022) P10019, arXiv:2210.09048, §3.1; M. Arratia, D. Britzger, O. Long, B. Nachman, Nucl. Instrum. Meth. A 1025 (2022) 166164.

[18] A. Jentsch, Z. Tu, C. Weiss, Phys. Rev. C 104 (2021) 065205, arXiv:2108.08314.

[19] Second EIC detector / IR-8 far-forward design: *Precision studies of QCD in the low energy domain of the EIC*, arXiv:2211.15746 (§VIII); BNL LDRD 23-050 closeout, arXiv:2602.04636.

[20] M. Pitt, PoS DIS2024 (2025) 259, arXiv:2409.02811; E.-C. Aschenauer et al., Phys. Rev. D 112 (2025) 036010, arXiv:2503.05908.

[21] C. Peng, *Superconducting nanowires at the ePIC far-forward: what a closer approach would buy the polarized-lithium tags*, Report 4 of this series (2026).

[22] W. Chang, E.-C. Aschenauer, A. Jentsch, A. Kumar, Z. Tu, Z. Yin, *Opportunities for imaging light nuclei with a second interaction region at the EIC*, Phys. Rev. D 113 (2026) 032018, arXiv:2511.05638.

[23] H1 Collaboration, Eur. Phys. J. C 72 (2012) 2074, arXiv:1203.4495.

[24] T. Lappi, Nucl. Phys. A 830 (2009) 403c, arXiv:0907.4588; H. Kowalski, T. Lappi, C. Marquet, R. Venugopalan, Phys. Rev. C 78 (2008) 045201, arXiv:0805.4071.

[25] I. Sick, arXiv:1505.06924.

[26] STAR Collaboration, Phys. Rev. Lett. 128 (2022) 122303, arXiv:2109.07625.

[27] L. R. Suelzle, M. R. Yearian, H. Crannell, Phys. Rev. 162 (1967) 992.

[28] G. C. Li, I. Sick, R. R. Whitney, M. R. Yearian, Nucl. Phys. A 162 (1971) 583.

[29] H. Mäntysaari, H. Roch, B. Schenke, C. Shen, W. Zhao, Phys. Rev. D 114 (2026) 014068, arXiv:2605.00454.

[30] H. Mäntysaari, P. Singh, Eur. Phys. J. C 85 (2025) 1449, arXiv:2411.14934.

[31] K. D. Veal, C. R. Brune, W. H. Geist et al., Phys. Rev. C 60 (1999) 064003.

[32] R. B. Wiringa, R. Schiavilla, Phys. Rev. Lett. 81 (1998) 4317, nucl-th/9807037.

[33] M. Bouwhuis et al. (NIKHEF), Phys. Rev. Lett. 82 (1999) 3755; D. Abbott et al. (JLab t20), Phys. Rev. Lett. 84 (2000) 5053; D. M. Nikolenko et al. (VEPP-3), Phys. Rev. Lett. 90 (2003) 072501.

[34] A. Airapetian et al. (HERMES), Phys. Rev. Lett. 95 (2005) 242001, hep-ex/0506018.

[35] N. N. Nikolaev, W. Schäfer, Phys. Lett. B 398 (1997) 245.

[36] K. Bora, R. L. Jaffe, Phys. Rev. D 57 (1998) 6906, hep-ph/9711323.

[37] J. Edelmann, G. Piller, W. Weise, Phys. Rev. C 57 (1998) 3392.

[38] Y. Hatta, B.-W. Xiao, F. Yuan, Phys. Rev. Lett. 116 (2016) 202301.

[39] I. Angeli, K. P. Marinova, At. Data Nucl. Data Tables 99 (2013) 69; N. J. Stone, At. Data Nucl. Data Tables 111–112 (2016) 1.

[40] D. R. Tilley et al., *Energy levels of light nuclei A = 5, 6, 7*, Nucl. Phys. A 708 (2002) 3 (TUNL).

[41] LEPS Collaboration, coherent ϕ photoproduction on ^4He , arXiv:1711.01095.

[42] W. Chang et al., Phys. Rev. D 104 (2021) 114030, arXiv:2108.01694.

[43] ZEUS Collaboration, Nucl. Phys. B 816 (2009) 1, arXiv:0812.2003 (leading-proton azimuthal asymmetries A_{LT} , A_{TT}).

Appendix A · Revision history

DATE	REVISION
2026-08-10	First version (development run 5): money plots 5 and 6 with the moment-constrained $\Delta(x,Q^2)$ and separate one-year / ten-year projections, and the physics note with a verified bibliography; first audit round the same day.
2026-08-11	Second audit round (formulas against the literature, implementation, references, extraction logic); money plot 7, the extracted Δ as structure-function data points, added.
2026-08-17	Dedicated fact checks of the Sather–Schmidt moment usage and of the far-forward detection chain; phase-space figure with the analysis binning; display mathematics typeset at build time.
2026-08-24	The anchored coefficient's convention ($2a_2$); the reconstruction-chain report created as the reconstructed-level companion.
2026-08-25	The spin-state-sorted ratio as the estimator of the measurement; the near-beam cut stated as angular; the b_1 , b_2 sign convention; the inversion corrected; the $O(\gamma^2)$ b_1 leakage; far-forward parameters updated.
2026-08-26	Development run 8: the PYTHIA 8 hadronic final state; the Roman-Pot aperture measured in the ePIC geometry; the published R1998 fit behind an <code>r_func</code> hook; the forward-folded bin-centering alternative.

2026-08-27 Rewritten as a paper on the γ -matched beam energies (${}^6\text{Li}$ at 40.8 / 99.5 / 137.5 GeV/u, plans/10). All numbers restated from the current scripts: sweet spots at $x = 0.011\text{--}0.14$, amplitudes $(0.44\text{--}0.95)\times 10^{-2}$ against $\delta A = (1.4\text{--}4.5)\times 10^{-4}$, the reconstructed-level Table 1, the R-model shifts (+18, +19, +8, -4%), the bin-centering scan, 1.2×10^8 coherent events and 1.7×10^7 tagged for the 0.20 GeV envelope. The coherent channel restated against the Yellow Report divergences and the measured pot aperture (Table 2): it does not survive at IP6 with any published optics, and the figures are presented for a tagging optics or IR-8 rather than as an IP6 projection; the pre-2026-08-27 6R and measured-aperture results, computed at the superseded rigidity-scaled energies (2.7×10^6 tagged at 50 GeV/u; the low-energy survival at 20.5 GeV/u), are withdrawn and replaced by the analysis closure at a 0.73 mrad approach. Assumptions condensed to Table 4; references numbered in citation order.

Generated from the PolarizedLithiumSim repository: `evgen/scripts/money_cos2phi.py`, `money_cos2phi_coherent.py`, `money_delta_extraction.py`, `phase_space_bins.py`; reconstructed level `money_cos2phi_reco.py` (with `--y-source hfs`, `--unfold-scan`, `--syst-scan`), `hfs_resolution.py`, `coherent_optics_scan.py`, `nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py`; models in `polli_fastsim/delta_models.py` and `evgen/polligen/coherent.py`; assembled by `reports/build_report.py`. Every command is in `docs/reproduction_manual.md`. All rates use scenario inputs — bands, not predictions; external numbers are verified against primary sources. Target venue: INT programme on polarized ion beams at the EIC, March 22 – April 2, 2027.